Algebra Essentials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating expressions

If $x = 4$, evaluate $3x + 7$.

$3(4) + 7 = 12 + 7 = 19$.

2 [Foundational] • Target: Evaluating expressions

If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.

$5^2 - 3(2) = 25 - 6 = 19$.

3 [Foundational] • Target: Evaluating expressions

If $x = -3$, evaluate $2x^2 + 5x$.

$2(-3)^2 + 5(-3) = 2(9) - 15 = 18 - 15 = 3$.

4 [Foundational] • Target: Evaluating expressions

If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.

$\frac{6 - (-2)}{-2 + 6} = \frac{8}{4} = 2$.

5 [Foundational] • Target: Solving linear equations

Solve $2x + 5 = 17$.

$2x = 12$, so $x = 6$.

6 [Foundational] • Target: Solving linear equations

Solve $5x - 8 = 27$.

$5x = 35$, so $x = 7$.

7 [Foundational] • Target: Solving linear equations

Solve $3(x - 4) = 21$.

$x - 4 = 7$, so $x = 11$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Solving linear equations

Solve $\frac{x}{4} + 3 = 10$.

$\frac{x}{4} = 7$, so $x = 28$.

9 [Intermediate] • Target: Solving linear equations

Solve $7x + 3 = 4x + 18$.

$7x - 4x = 18 - 3$, so $3x = 15$, and $x = 5$.

10 [Intermediate] • Target: Expanding and simplifying

Expand $(x + 3)(x + 5)$.

$x^2 + (3 + 5)x + 3 \times 5 = x^2 + 8x + 15$.

11 [Intermediate] • Target: Expanding and simplifying

Expand $(x - 4)(x + 6)$.

$x^2 + (-4 + 6)x + (-4)(6) = x^2 + 2x - 24$.

12 [Intermediate] • Target: Expanding and simplifying

Expand $(2x + 1)(x - 3)$.

$2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$.

13 [Intermediate] • Target: Expanding and simplifying

Simplify $3(2x - 5) - 2(x - 4)$.

$6x - 15 - 2x + 8 = 4x - 7$.

14 [Intermediate] • Target: Factoring

Factor $x^2 + 9x + 20$.

Find two numbers with sum 9 and product 20: 4 and 5. $(x + 4)(x + 5)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Factoring • Cross-Lesson: 4.01 Geometry & Measurement

Factor $x^2 - 2x - 15$.

Find two numbers with sum -2 and product -15: -5 and 3. $(x - 5)(x + 3)$.

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16 [Challenging] • Target: Factoring • Cross-Lesson: 5.01 Quantitative Comparisons

Factor $x^2 - 49$.

Difference of squares: $(x - 7)(x + 7)$.

17 [Challenging] • Target: Factoring • Cross-Lesson: 6.01 Sequences & Patterns

Factor $6x + 15$.

Common factor 3: $3(2x + 5)$.

18 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 7.01 Data Analysis & Probability

Solve $x^2 + 5x + 6 = 0$.

Factor: $(x + 2)(x + 3) = 0$, so $x = -2$ or $x = -3$.

19 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 8.01 Word Problems & Reasoning

Solve $x^2 - 7x + 12 = 0$.

Factor: $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.

20 [Challenging] • Target: Solving quadratic equations by factoring • Cross-Lesson: 1.01 Arithmetic & Number Sense

Solve $x^2 - 16 = 0$.

Difference of squares: $(x - 4)(x + 4) = 0$, so $x = 4$ or $x = -4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving quadratic equations

The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?

a. x = 7 only

b. x = 49

c. **x = 7 or x = -7 ✓**

d. x = -7 only

The curve crosses the x-axis at $x = -7$ and $x = 7$, matching $x^2 - 49 = 0$.

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22 [Foundational] • Target: Systems of two equations

The graph shows two lines. What is their point of intersection?

a. (1, 4)

b. (0, 5)

c. **(2, 3)** ✓

d. (3, 2)

The two lines $y = x + 1$ and $y = -x + 5$ cross where both are true: at $(2, 3)$.

23 [Foundational] • Target: Inequalities

The number line shows the solution set of an inequality. Which inequality matches?

a. $x \geq 4$

b. $x \leq 4$

c. $x > 4$

d. $x < 4$ ✓

An open circle at 4 with shading to the left means $x < 4$.

24 [Foundational] • Target: Reading function graphs

The graph shows a line. What is its slope?

a. $\frac{1}{2}$

b. 2 ✓

c. -2

d. -1

The line rises 2 for every 1 across, e.g. from $(0, -1)$ to $(1, 1)$: slope $= 2$.

25 [Foundational] • Target: Reading function graphs

What is the y-intercept of the line shown?

a. 4 ✓

b. 0

c. -4

d. 1

The line crosses the y-axis at $(0, 4)$.

26 [Foundational] • Target: Reading function graphs

What is the equation of the line shown?

a. $y = 3x + 1$

b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$ ✓

The line passes through the origin and $(1, 3)$: slope 3, no intercept, so $y = 3x$.

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27 [Foundational] • Target: Reading function graphs

Find the slope of the line through the two marked points.

a. 2 ✓

b. $\frac{1}{2}$

c. -2

d. 4

Slope $= \frac{6-2}{3-1} = \frac{4}{2} = 2$.

28 [Foundational] • Target: Substitution into formulas

The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w.

a. 8

b. 6 ✓

c. 384

d. 40

$48 = 8w \Rightarrow w = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Word problems

The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?

a. Student B

b. Student D

c. **Student C** ✓

d. Student A

$70 + 12 = 82$, which is Student C's score.

30 [Intermediate] • Target: Evaluating expressions

If $x = 6$, evaluate $4x - 9$.

a. 33

b. 24

c. -15

d. 15 ✓

$4(6) - 9 = 24 - 9 = 15$.

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31 [Intermediate] • Target: Evaluating expressions

If $a = -2$ and $b = 3$, evaluate $a^2 + 2ab$.

a. -16

b. 8

c. -4

d. -8 ✓

$a^2 = 4$, $2ab = 2(-2)(3) = -12$. $4 + (-12) = -8$.

32 [Intermediate] • Target: Evaluating expressions

If $x = 5$ and $y = -1$, evaluate $(x + y)^2 - xy$.

a. 16

b. 11

c. 21 ✓

d. -5

$(x + y)^2 = 4^2 = 16$; $xy = -5$. $16 - (-5) = 21$.

33 [Intermediate] • Target: Evaluating expressions

If $p = 3$, evaluate $2p^3 - 5p$.

a. 49

b. 39 ✓

c. 33

d. 69

$2(27) - 15 = 54 - 15 = 39$.

34 [Intermediate] • Target: Solving linear equations

Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$ ✓

$4x + 8 = 3x + 15 \Rightarrow x = 7$.

35 [Intermediate] • Target: Solving linear equations

Solve $\frac{2x - 1}{3} = 5$.

a. $x = 4$

b. $x = 7$

c. $x = 16$

d. $x = 8$ ✓

$2x - 1 = 15 \Rightarrow 2x = 16 \Rightarrow x = 8$.

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36 [Intermediate] • Target: Solving linear equations

Solve $6 - 2x = 3x - 9$.

a. $x = -3$

b. $x = 3$ ✓

c. $x = 15$

d. $x = 1$

$15 = 5x \Rightarrow x = 3$.

37 [Intermediate] • Target: Expanding and simplifying

Expand $(x + 7)^2$.

a. $x^2 + 7x + 49$

b. $x^2 + 14x + 14$

c. $x^2 + 14x + 49$ ✓

d. $x^2 + 49$

$(x + 7)^2 = x^2 + 2(7)x + 49 = x^2 + 14x + 49$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 7.01 Data Analysis & Probability

Expand $(x - 5)^2$.

a. $x^2 + 10x + 25$

b. $x^2 - 5x + 25$

c. $x^2 - 10x + 25$ ✓

d. $x^2 - 25$

$(x - 5)^2 = x^2 - 10x + 25$.

39 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 8.01 Word Problems & Reasoning

Expand $(3x - 2)(3x + 2)$.

a. $9x^2 + 4$

b. $9x^2 - 12x - 4$

c. $9x^2 - 4$ ✓

d. $6x^2 - 4$

Difference of squares: $(3x)^2 - 2^2 = 9x^2 - 4$.

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40 [Challenging] • Target: Expanding and simplifying • Cross-Lesson: 1.01 Arithmetic & Number Sense

Simplify $2(x+3)-3(x-1)$.

a. $5x+9$

b. $-x+9$ ✓

c. $x+9$

d. $-x+3$

$2x+6-3x+3=-x+9$.

41 [Challenging] • Target: Factoring • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Factor $3x^2+12x$.

a. $3(x^2+4x)$

b. $x(3x+12)$

c. $3x(x+12)$

d. $3x(x+4)$ ✓

GCF $=3x$: $3x^2+12x=3x(x+4)$.

42 [Challenging] • Target: Factoring • Cross-Lesson: 4.01 Geometry & Measurement

Factor $x^2-11x+30$.

a. $(x-5)(x-6)$ ✓

b. $(x-2)(x-15)$

c. $(x+5)(x+6)$

d. $(x-3)(x-10)$

Need two numbers multiplying to 30, adding to -11: $-5,\ -6$.

43 [Challenging] • Target: Factoring • Cross-Lesson: 5.01 Quantitative Comparisons

Factor $x^2+3x-18$.

a. $(x+2)(x-9)$

b. $(x-6)(x+3)$

c. $(x+9)(x-2)$

d. $(x+6)(x-3)$ ✓

Need two numbers multiplying to -18, adding to 3: $6,\ -3$.

44 [Challenging] • Target: Solving quadratic equations • Cross-Lesson: 6.01 Sequences & Patterns

Solve $2x^2-8x=0$.

a. x = 8

b. x = 2 or x = 4

c. **x = 0 or x = 4** ✓

d. x = 0 or x = -4

$2x(x-4)=0 \Rightarrow x=0$ or $x=4$.

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45 [Challenging] • Target: Systems of two equations • Cross-Lesson: 7.01 Data Analysis & Probability

Solve the system: $x + 2y = 8$, $x - y = 2$.

a. x = 2, y = 3

b. x = 6, y = 1

c. **x = 4, y = 2 ✓**

d. x = 8, y = 0

Subtracting: $3y = 6 \Rightarrow y = 2$, then $x = 4$.

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